

HMM-GDAN: Hybrid multi-view and multi-scale graph duplex-attention networks for drug response prediction in cancer

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ABSTRACT

Precision medicine is devoted to discovering personalized therapy for complex and difficult diseases like cancer. Many machine learning approaches have been developed for drug response prediction towards precision medicine. Notwithstanding, genetic profiles based multi-view graph learning schemes have not yet been explored for drug response prediction in previous works. Furthermore, multi-scale latent feature fusion is not considered sufficiently in the existing frameworks of graph neural networks (GNNs). Previous works on drug response prediction mainly depend on sequence data or single-view graph data. In this paper, we propose to construct multi-view graph by means of multi-omics data and STRING protein–protein association data, and develop a new architecture of GNNs for drug response prediction in cancer. Specifically, we propose hybrid multi-view and multi-scale graph duplex-attention networks (HMM-GDAN), in which both multi-view self-attention mechanism and view-level attention mechanism are devised to capture the complementary information of views and emphasize on the importance of each view collaboratively, and rich multi-scale features are constructed and integrated to further form high-level representations for better prediction. **Experiments on GDSC2 dataset verify the superiority of the proposed HMM-GDAN when compared with state-of-the-art baselines.** The effectiveness of multi-view and multi-scale strategies is demonstrated by the ablation study.

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1. Introduction

Cancer is a difficult disease to treat in the world. Precision medicine in cancer analyses tumours at the level of molecules and might suggest personalized treatment. Genetic profiles based precision medicine is not perfect (Friedman, Letai, Fisher, & Flaherty, 2015) and casual effect may be incorrectly revealed. Measurement error of the instrument may also have impact on the genetic analysis. Therefore, large-scale drug screening experiments are done and related computational models are established to predict the cancer patients' response to target-specific drugs. Many anti-cancer drugs against a set of cancer cell lines have been screened for recent years. For example, two large-scale pharmacogenomic datasets have been publicly available, namely, the Cancer Cell Line Encyclopedia (CCLE) (Barretina, Caponigro, Stransky, et al., 2012) and the Genomics of Drug Sensitivity in Cancer (GDSC) (Garnett, Edelman, Heidorn, et al., 2012).

Classical machine learning algorithms, including support vector machine (SVM) (Hearst, Dumais, Osuna, Platt, & Scholkopf, 1998) and random forest (RF) (Qi, 2012), have been successfully applied for drug response prediction. These methods may

suffer from overfitting for high-dimensional input data and rely on effective feature selection. Deep learning is a brain-inspired method to learn rich representations from (regular) data, such as text, image and video. Existing works, such as tCNCN (Liu, Li, Li, & Leung, 2019) and DeepCDR (Liu, Hu, Jiang, & Zhou, 2020), have considered convolutional neural networks (CNNs) or fully connected networks (FCNs) for sequence data of cell lines. Nevertheless, it cannot deal with irregular data like molecular graph. To deal with this problem, graph neural networks (GNNs) have been proposed in Kipf and Welling (2016) and become one of the most active topics in recent years. GNNs can learn rich and hierarchical representation of nodes for graph-structural data. GNNs have been shown to be powerful in social networks (Fan et al., 2019), traffic forecasting (Rico, Barateiro, & Oliveira, 2021), materials discovery (Jiang, Yang, Guo, et al., 2021) and so on. Due to the natural graph structure for molecules in drugs, researchers have made an attempt to apply techniques of GNNs to the field of drug response prediction. Liu et al. proposed hybrid graph convolutional networks (Liu et al., 2020) for drug response prediction in cancer. Zhu et al. proposed protein–protein association-based twin graph neural networks (Zhu et al., 2022) to extract features for drugs and cell lines and use the trick of similarity augmentation (SA) (Zhu et al., 2022) to enhance the prediction performance.

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Although these methods have made success in the task of drug response prediction, they do not explore the multi-view approaches which exploit useful information from different views. As a matter of fact, multi-view approach is suitable for the drug response prediction problem because the training data of cell lines can be expressed as multi-view graph data with appropriate construction methods of multi-view graph. In light of this, we propose multi-view graph learning schemes. Specifically, we provide the construction method of multi-view graph for cell lines on the basis of multi-omics data and protein–protein association data. Generally speaking, multi-omics data includes the genome, epigenome, transcriptome, metabolome, and other molecular data. Hence we can use each omics data to generate a view. Different views reflect on different information. In this way, these views induce different adjacent matrices. Except for multi-omics data, we also use the STRING dataset (i.e. protein–protein association data) to generate an additional view. This view is quite different from those views generated from multi-omics data. Hence the constructed views are heterogeneous and embody information of different types. Note that information in a single view is incomplete. By the proposed construction of multi-view graph, we have the chance to further leverage the complementary information among different views.

With this principle, we further propose hybrid multi-view and multi-scale graph duplex-attention networks (HMM-GDAN) for drug response prediction. The duplex-attention mechanism means joint multi-view self-attention mechanism and view-level mechanism. The multi-view self-attention captures the pairwise dependency between views and can be used to enhance representations in one view by leveraging complementary information from other views. To the best of our knowledge, it is the first time to propose multi-view self attention mechanism for multi-view GNNs. View-level attention mechanism emphasizes on the importance of each view from the global perspective. Multi-scale features are also constructed in different ways, such as concatenation, Hadamard product and projection, and fused in an organic way. Multi-scale feature means a feature is formed by the concatenation of multiple features with different dimensions. Hence, multi-scale feature is indeed a composite feature which contains multiple multi-dimensional features.

As summarization, contributions of this paper are as follows.

- It is the first time to consider multi-view graph learning schemes in the research of drug response prediction. To prepare for multi-view learning, we develop an effective construction method of multi-view graph by means of multi-omics data and protein–protein association data.
- Duplex-attention mechanism is devised for multi-view graph learning, which leverages the complementary information between views and captures the importance of each view collaboratively. This kind of mechanism can be easily inserted into multi-view learning methods and make feature representations more informative in each view.
- To implement an end-to-end multi-view graph learning scheme, we propose hybrid multi-view and multi-scale graph duplex-attention networks (HMM-GDAN) for drug response prediction, which leverage the useful multi-view information to learn better representations with multi-scale fusion.

2. Related works

2.1. GNNs

GNNs have drawn much attention by machine learning community and a large amount of works have emerged for recent

years. The related works mainly include GCN (Kipf & Welling, 2016), GAT (Veličković et al., 2017) and GIN (Xu, Hu, Leskovec, & Jegelka, 2018). GCN (Kipf & Welling, 2016) is induced by spectral graph convolution via graph Fourier transformation. It has elegant layer update expressions when approximate Chebyshev polynomials are applied. GAT (Veličković et al., 2017) uses node-level attention mechanism to the aggregation of neighbourhood's information with adaptive attention weights. GIN (Xu et al., 2018) is a graph isomorphic network which has been proven to be as powerful as Weisfeiler–Lehman test theoretically. GCN and GIN are jointly modified in this paper for extracting multi-scale features.

2.2. Attention mechanism

Attention mechanism (Chorowski, Bahdanau, Serdyuk, Cho, & Bengio, 2015) is initially proposed for deep neural networks in the task of image recognition. It can help deep neural networks pay much attention to the important areas in an image and learn better representations for classification or regression task. Then it is extended to GNNs (e.g. GAT Veličković et al., 2017) for learning the node's representation and brings the progress in performance improvement. Recently, multi-view graph attention mechanism is also proposed in Xie, Zhang, Gong, Tang, and Han (2020). This attention is essentially node-to-view format. It can enhance the representation of nodes in each view by blending representations of all views with attention coefficients.

2.3. Feature fusion

Multi-scale fusion (Jiang, Wang, Yi, et al., 2020) is an effective strategy to enrich the meaning of learnt representation. It remedies the drawback of single-scale feature that may leave out some detailed information. Multi-scale fusion has been successfully applied to object detection (Zhang, Liu, Piao, Yao, & Lu, 2021), drug–drug interaction prediction (Chen et al., 2021) and multivariate time series forecasting (Cui, Zheng et al., 2021). Multi-view feature fusion is also popular approach to generate rich representations from the perspective of multi-view learning. For example, Chen et al. develop an end-to-end graph convolutional network to perform feature fusion and graph fusion for multi-view learning in Chen et al. (2023). Except for multi-scale and multi-view fusion, high dimensional feature products (Hua, Rabusseau, & Tang, 2022) brings a different feature fusion way with the help of tensor decomposition, which results in high-order attribute interaction.

2.4. Drug response prediction

Traditional machine learning methods have been successfully applied in drug response prediction, which are summarized in Adam et al. (2020) systematically. Subsequently, deep learning methods emerge and are widely used for drug response prediction (Baptista, Ferreira, & Rocha, 2021). CDRscan (Chang, Park, Yang, et al., 2018) is a deep learning method which predicts anticancer drug responsiveness by means of large-scale drug screening assay data including human cancer cell lines and drugs. MOLI (Sharifi-Noghabi, Zolotareva, Collins, & Ester, 2019) considers a multi-omics integration scheme with deep neural networks for enhancing the prediction power. However, the drug data is usually depicted by graph structure (e.g. molecular graph). The drug response prediction is not well-studied with these deep learning-based methods. Therefore, GNNs serve as one of the best choices for learning representations. In GraphDRP (Nguyen, Nguyen, Nguyen, & Le, 2021), drugs were depicted with molecular graphs which reflect on the bonds among atoms and graph convolutional networks can directly act on them for classification and regression tasks.

3. Methodology

3.1. Problem setup

Let S_d and S_{cl} be the set of drugs and cell lines respectively. Drug response in cell lines are typically measured by the half maximal inhibitory concentration (IC50) (Garnett et al., 2012). Assume that there is a given drug response training set $\mathcal{E} = \{(dr_i, cl_j, y_{ij})\}_{1 \leq i \leq M_d, 1 \leq j \leq M_{cl}}$, where M_d is the number of drugs, M_{cl} is the number of cell lines and y_{ij} is the log-normalized half maximal inhibitory concentration (ln(IC50)) values for the pair of drug i and cell line j . Multi-omics data with respect to cancer-related genes for cell lines is also provided. The aim is to establish a regression model $f: S_d \times S_{cl} \rightarrow \mathbb{R}$ to predict ln(IC50) values for drug-cell line pairs in the test set.

With this aim, we first formulate the construction scheme of multi-view graph, which is a vital important step towards making use of multi-view GNNs. Then we propose multi-view duplex-attention networks for cell lines and multi-scale graph neural networks for drugs so that graph-level features can be generated and used for subsequent regression model of drug response prediction in a hybrid way.

3.2. The construction of multi-view graph

In standard Statistics, Pearson correlation coefficient (PCC) measures the linear correlation (dependence) between two random variables Y and Z . Mathematically, it is calculated as

$$\rho_{YZ} = \frac{\text{Cov}(Y, Z)}{\sigma_Y \sigma_Z}, \quad (1)$$

where $\text{Cov}(\cdot, \cdot)$ is covariance operator and σ stands for standard deviation.

Given i.i.d. samples $\{Y_i\}_{i=1}^M$ and $\{Z_i\}_{i=1}^M$, PCC can be estimated as

$$\rho_{YZ} \approx \frac{\sum_{i=1}^M (Y_i - \bar{Y})(Z_i - \bar{Z})}{\sqrt{\sum_{i=1}^M (Y_i - \bar{Y})^2} \sqrt{\sum_{i=1}^M (Z_i - \bar{Z})^2}}, \quad (2)$$

where both $\bar{Y}$ and $\bar{Z}$ represent mean value, namely

$$\bar{Y} = \frac{1}{M} \sum_{i=1}^M Y_i, \bar{Z} = \frac{1}{M} \sum_{i=1}^M Z_i. \quad (3)$$

It is well-known that $\rho_{YZ} \in [-1, 1]$.

Given (K – 1)-omics data (Barretina et al., 2012) and STRING dataset (Szklarczyk, Gable, Lyon, et al., 2019), we can construct multi-view graph for cell lines as follows. The first $K - 1$ views are formed with the help of multi-omics data and the last view depends on STRING dataset.

For given $K - 1$ omics data, the third-order tensor $C \in \mathbb{R}^{N_{cl} \times N_g \times (K-1)}$ is formed by folding operation, where N_{cl} is the number of cell lines, N_g is the number of selected cancer-related genes and K is the number of views. We propose two usages for this tensor. That is, it can produce node features and multi-view graph.

Node feature matrix for i th cell line is directly obtained by taking

$$X_c^i = C(i, :, :) \in \mathbb{R}^{N_g \times (K-1)}, \quad (4)$$

where N_g is the number of genes, $K - 1$ is the number of omics and $i = 1, 2, \dots, N_{cl}$.

Let

$$C(:, :, v) \in \mathbb{R}^{N_{cl} \times N_g} \quad (5)$$

be the v th slice of tensor C , which stands for the v th omics data. We show the meaning of node feature matrices and multi-omics in Fig. 1 for intuition.

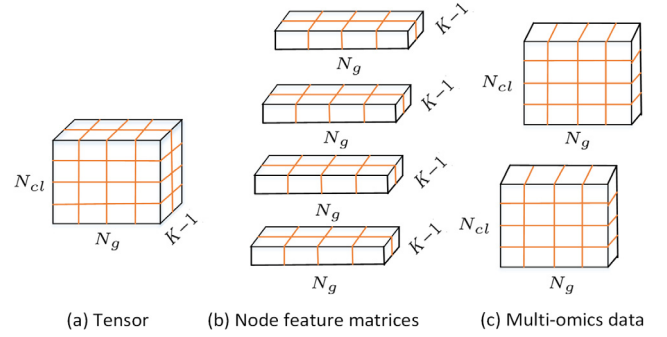


Fig. 1. The illustration of tensor C and two induced data.

Let

$$Y_i^{(v)} = [Y_{i,1}^{(v)}, Y_{i,2}^{(v)}, \dots, Y_{i,N_{cl}}^{(v)}]^T = C(:, i, v) \in \mathbb{R}^{N_{cl}} \quad (6)$$

be the observation vector of gene i in the v th slice of tensor C , whose components correspond to the observation in cell lines.

For $v \in \{1, 2, \dots, K - 1\}$, adjacency matrix $E^{(v)} \in \{0, 1\}^{N_g \times N_g}$ of v th view is prescribed as

$$E_{ij}^{(v)} = \begin{cases} 1, & |\rho_{ij}^{(v)}| > \tau^{(v)}; \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where $\tau^{(v)}$ is a predefined threshold and

$$\rho_{ij}^{(v)} = \frac{\sum_{q=1}^{N_g} (Y_{i,q}^{(v)} - \bar{Y}_i^{(v)})(Y_{j,q}^{(v)} - \bar{Y}_j^{(v)})}{\sqrt{\sum_{q=1}^{N_g} (Y_{i,q}^{(v)} - \bar{Y}_i^{(v)})^2} \sqrt{\sum_{q=1}^{N_g} (Y_{j,q}^{(v)} - \bar{Y}_j^{(v)})^2}} \quad (8)$$

in which

$$\bar{Y}_i^{(v)} = \frac{1}{N_g} \sum_{q=1}^{N_g} Y_{i,q}^{(v)}, \bar{Y}_j^{(v)} = \frac{1}{N_g} \sum_{q=1}^{N_g} Y_{j,q}^{(v)}. \quad (9)$$

These $K - 1$ views are established by means of multi-omics data. At this end, the multi-omics data have been used to produce node features and construct multi-view graph.

For STRING dataset (Szklarczyk et al., 2019), adjacency matrix is established as follows. Namely, for $v = K$, adjacency matrix $E^{(v)} \in \mathbb{R}^{N_g \times N_g}$ is prescribed as

$$E_{ij}^{(v)} = \begin{cases} 1, & s_{ij}^{(v)} > \tau^{(v)}; \\ 0, & \text{otherwise,} \end{cases} \quad (10)$$

where $\tau^{(v)}$ is a predefined threshold and $s_{ij}^{(v)}$ is the interaction score between gene i and j in STRING (Szklarczyk et al., 2019).

In summary, we have constructed multi-view graph $G_c = \langle V, \{E^{(v)}\}_{v=1}^K \rangle$ for cell lines, where V denoted the set of selected genes with respect to cancer.

3.3. Multi-view graph duplex-attention networks for cell lines

For the well-established multi-view graph of cell lines in Section 4.1, multi-view node-level embeddings can be generated as follows. Assume that $X_c \in \mathbb{R}^{N_g \times K}$ be the feature matrix of a cell line. For v th view,

$$H^{(v)} = \text{GNN}_c(X_c, E^{(v)}) \in \mathbb{R}^{N_g \times d_c}, \quad (11)$$

where N_g is the number of genes, d_c is the hidden dimension, and GNN_c is the chosen architecture of graph neural networks for cell lines by users. In the experiments, we will point out the specific choice. Now we propose duplex-attention mechanism (See Fig. 2 for intuition) which includes multi-view self-attention mechanism and view-level attention mechanism.

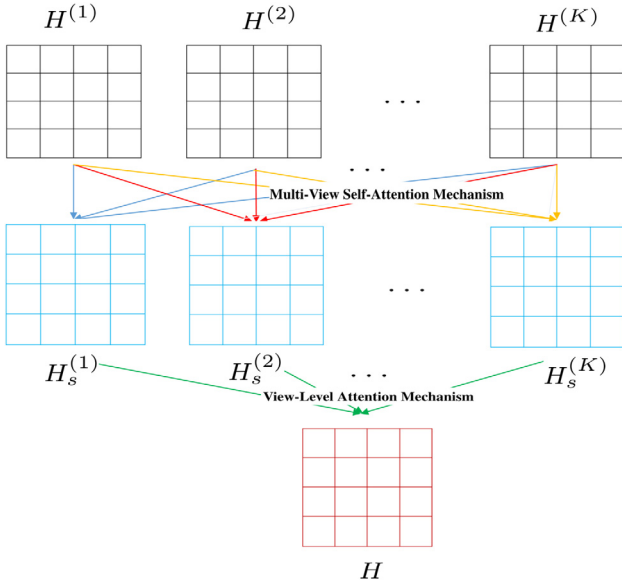


Fig. 2. The illustration of duplex-attention mechanism. It is the coupling of multi-view self-attention mechanism and view-level attention. The former captures the complementary information between views. The latter underlines the importance of single view.

3.3.1. Multi-view self-attention mechanism

First, we employ a shared parametric vector $\theta \in \mathbb{R}^{d_c}$ for representations of all views, i.e.

$$\Theta = [H^{(1)}\theta, H^{(2)}\theta, \dots, H^{(K)}\theta] \in \mathbb{R}^{N_g \times K}, \quad (12)$$

where N_g is the number of genes and K is the number of views. Each column of $\Theta = [\Theta_1, \Theta_2, \dots, \Theta_K]$ can be regarded as the new representation of each view. In this way, the relationship among views is established by the introduction of sharing vector θ . Based on these representations of views, the similarity of view i and j can be characterized by

$$\begin{aligned} \bar{\alpha}_{ij} &= \text{LeakyReLU}(\cos(\Theta_i, \Theta_j), \beta) \\ &= \text{LeakyReLU}\left(\frac{\langle \Theta_i, \Theta_j \rangle}{\|\Theta_i\|_2 \|\Theta_j\|_2}, \beta\right), \end{aligned} \quad (13)$$

where LeakyReLU is

$$\text{LeakyReLU}(x, \beta) = \begin{cases} x, & x > 0; \\ -\beta x, & x \leq 0. \end{cases} \quad (14)$$

Here $\beta \geq 0$ is specified by users. Usually, we set $\beta = 0.1$ in experiments.

Multi-view self-attention coefficient between view i and j is then calculated as

$$\alpha_{ij} = \frac{\exp(\bar{\alpha}_{ij})}{\sum_{j=1}^K \exp(\bar{\alpha}_{ij})}, \quad (15)$$

where K is the number of views and $\bar{\alpha}_{ij}$ comes from Eq. (13). It is exactly the softmax normalization of Eq. (13) such that $\sum_{j=1}^K \alpha_{ij} = 1, \forall i = 1, 2, \dots, K$.

In view of the proposed multi-view self-attention mechanism, we can update the original $\{H^{(v)}\}_{v=1}^K$ as

$$H_s^{(v)} = \sum_{j=1}^K \alpha_{vj} H^{(j)}, \quad (16)$$

where $v \in \{1, 2, \dots, K\}$ and α_{vj} denote the attention weight of v th view with respect to j th view. Therefore, the complementary information under different views is sufficiently leveraged. To our

knowledge, existing works on multi-view graph neural networks, such as Cui, Du, Wang, Kou and Xue (2021) and Xie et al. (2020), only consider the node-to-view attention and do not explore this complementary knowledge among different views. We are the first to propose multi-view self-attention mechanism.

3.3.2. View-level attention mechanism

Let H be the concatenation of nodes' representation after using multi-view self-attention mechanism, i.e.

$$H_s = [H_s^{(1)}, H_s^{(2)}, \dots, H_s^{(K)}] \in \mathbb{R}^{N_g \times Kd_c}, \quad (17)$$

where N_g is the number of genes, K denotes the number of views, and d_c is the hidden dimension. Then we employ a group of parametric vectors $\{z^{(v)}\}_{v=1}^K$ to generate view weights

$$\bar{w}^{(v)} = \text{Avg}(H_s z^{(v)} + b) \quad (18)$$

for $v \in \{1, 2, \dots, K\}$, where K is the number of views, b is the bias and Avg denotes the average operator with

$$\text{Avg}(x) = \frac{1}{N_g} \sum_{i=1}^{N_g} x_i \quad (19)$$

for any $x \in \mathbb{R}^{N_g}$.

View-level attention coefficient is then the softmax normalization as

$$w^{(v)} = \frac{\exp(\bar{w}^{(v)})}{\sum_{v=1}^K \exp(\bar{w}^{(v)})}, \quad (20)$$

where $v \in \{1, 2, \dots, K\}$ and $\bar{w}^{(v)}$ comes from Eq. (18). It stands for the importance of the v th view.

Based on the proposed view-level attention mechanism, the integrated node representation of all K views is calculated as

$$H = \sum_{v=1}^K w^{(v)} H_s^{(v)}, \quad (21)$$

where $w^{(v)}$ denotes the importance of v th view and K is the number of views.

By multi-layer perceptron (MLP) and READOUT operation, we get the following graph-level feature

$$\mathcal{H} = \text{READOUT}(\text{MLP}(H)). \quad (22)$$

In experiments, READOUT operator in Eq. (22) is specified as concatenation by row vectors.

3.4. Multi-scale graph neural networks for drugs

We propose modified GIN-GCN architecture of multi-scale graph neural networks for drugs. Given drug features $X_d \in \mathbb{R}^{N_d \times m}$ and adjacency matrix $E_d \in \mathbb{R}^{N_d \times N_d}$, the aim is to construct a graph neural network GNN_d to extract features

$$J = GNN_d(X_d, E_d). \quad (23)$$

We propose the following modified GIN-GCN architecture as GNN_d . Let $\hat{E}_d = E_d + I$, where I is the identity matrix. Besides, let $\hat{D}_d$ be the diagonal matrix whose diagonal entry is $\hat{D}_{d,ii} = \sum_{j=1}^{N_d} \hat{E}_{d,ij}$. Then we set $\tilde{E}_d = \hat{D}_d^{-\frac{1}{2}} \hat{E}_d \hat{D}_d^{-\frac{1}{2}}$. By the standard GCN (Kipf & Welling, 2016), we get two-layer representations

$$X_d^{(i)} = \text{ReLU}(\tilde{E}_d X_d W_d^{(i-1)}) \in \mathbb{R}^{N_d \times m_i}, i = 1, 2. \quad (24)$$

The multi-scale features of these two layers are fused as

$$J_{mGCN} = [X_d^{(1)}, X_d^{(2)}, X_d^{(1)} + X_d^{(2)}, X_d^{(1)} \odot X_d^{(2)}], \quad (25)$$

where $\odot$ denotes the Hadamard product.

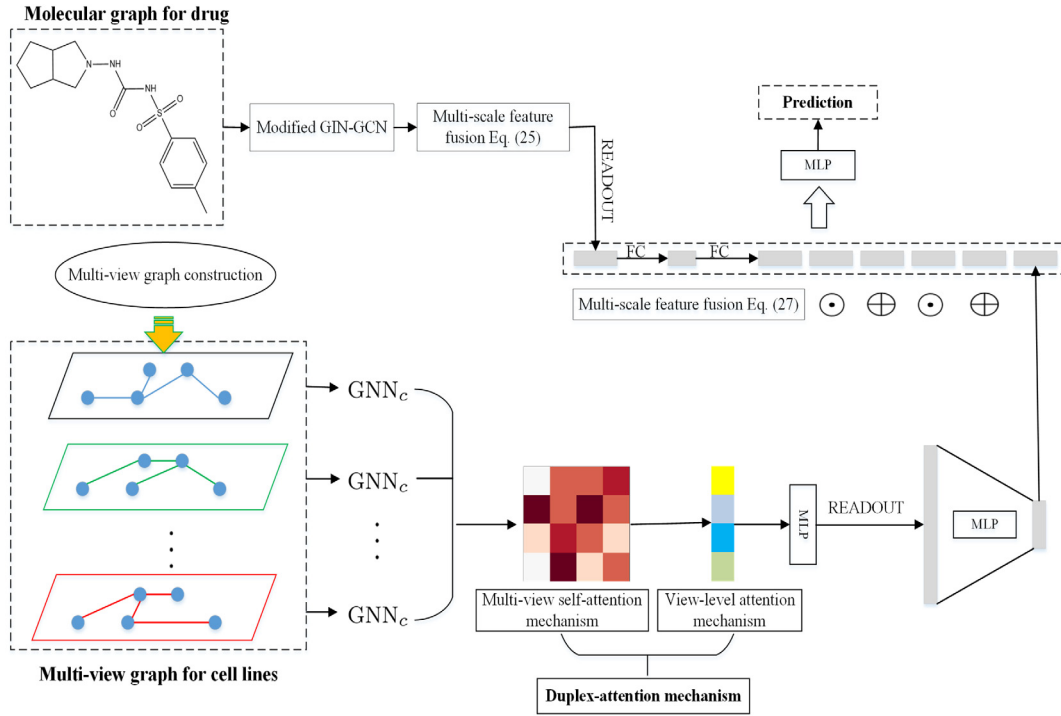


Fig. 3. The illustration of the proposed HMM-GDAN. It includes two basic branches: multi-view duplex-attention based graph neural networks for cell lines and multi-scale graph neural networks for drugs. Multi-view graph for cell lines is needed to be constructed from the scratch. Duplex-attention mechanism is a crucial module in the networks, which helps the mining of the useful knowledge hidden in all views. $\oplus$ and $\odot$ denote element-wise addition and product operation respectively.

Recall from Xu et al. (2018) that the layer update of GIN is formulated as

$$h_v^{(k)} = \text{MLP}^{(k)}(h_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)}), \quad (26)$$

where $\mathcal{N}(v)$ denotes the neighbours of node v . Generally, we take three layers in experiments, i.e. $H_d^{(1)}$, $H_d^{(2)}$, $H_d^{(3)}$. Fuse these representations as

$$J_{mGIN} = [H_d^{(1)}, H_d^{(2)}, H_d^{(3)}, \sum_{k=1}^3 H_d^{(k)}, H_d^{(1)} \odot H_d^{(2)} \odot H_d^{(3)}]. \quad (27)$$

Combining Eq. (25) with Eq. (27), we get the result of multi-scale feature fusion, i.e.

$$J = [J_{mGIN}, J_{mGCN}]. \quad (28)$$

By READOUT operation, we further have the following graph-level feature

$$\mathcal{L} = \text{READOUT}(J). \quad (29)$$

In experiments, READOUT operator in Eq. (29) is set as global max pooling.

3.5. Regression

Let $\mathcal{H}_i$ and $\mathcal{L}_j$ be the graph-level representations of i th drug and j th cell line in the training set. Here $\mathcal{H}_i$ is generated by multi-view graph duplex-attention mechanism and $\mathcal{L}_j$ is obtained by multi-scale graph neural networks. Assume that $\{\text{MLP}_i\}_{i=0}^4$ are five candidate MLPs.

Now the aim is to establish a regression model

$$\tilde{y}_{ij} = \mathcal{F}(\mathcal{H}_i, \mathcal{L}_j), \quad (30)$$

where $\mathcal{H}_i$ is the representation of i th cell line, $\mathcal{L}_j$ is the representation of j th cell line, and $\mathcal{F}$ is a regressor. In what follows, we elaborate on the construction of our regressor.

First, we use two MLPs to transform the graph-level features h_i and l_j into the same subspace, i.e.

$$h_i = \text{MLP}_0(\mathcal{H}_i) \in \mathbb{R}^D \quad (31)$$

and

$$l_j = \text{MLP}_1(\mathcal{L}_j) \in \mathbb{R}^D. \quad (32)$$

Then we project h_i into q -dimensional subspace and the reconstruct it back, namely,

$$h_{1,i} = \text{MLP}_2(h_i) \in \mathbb{R}^q, \quad (33)$$

$$h_{2,i} = \text{MLP}_3(h_{1,i}) \in \mathbb{R}^D. \quad (34)$$

The result of multi-scale feature fusion is

$$\Gamma_{ij} = [h_i; l_j; h_{1,i}; h_{2,i}; h_i + l_j; h_i \odot l_j; h_{2,i} + l_j; h_{2,i} \odot l_j] \in \mathbb{R}^{\bar{h}}, \quad (35)$$

where $\bar{h} = 7D + q$, ‘ $;$ ’ stands for concatenation along columns and $\odot$ denotes Hadamard product. Further, we do regression on this fused representation as

$$\tilde{y}_{ij} = \text{MLP}_4(\Gamma_{ij}). \quad (36)$$

Finally, the loss function is devised as

$$\mathcal{Q} = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} L(\tilde{y}_{ij}, y_{ij}) + \lambda \|\vartheta\|^2, \quad (37)$$

where $\lambda > 0$ is a trade-off parameter, ϑ is the overall network parameter vector and L is a loss function. L is usually set as square error function.

In summary, we display the complete pipeline of the proposed method in Fig. 3.

3.6. Complexity analysis

The complexity mainly depends the operation of matrix multiplication. In Eq. (12), K the matrix–vector multiplications need

Table 1
Details for feature configuration.

Attribute	Encoding	Length
Atom	One-hot encoding	43
Degree	One-hot encoding	11
Implicit valance	One-hot encoding	7
Formal charge	Integer encoding	1
Radical electrons	Integer encoding	1
Hybridization	One-hot encoding/zero	5
Aromatic	Binary encoding	1
Hydrogens	One-hot encoding	5
Ring	Binary encoding	1
Chirality	One-hot encoding/zero	2

$\mathcal{O}(N_g d_c K)$ complexity. In Eq. (13), the inner product can be implemented by matrix–matrix multiplication with $\mathcal{O}(K^2 N_g)$ complexity. Eq. (18) includes K matrix–vector multiplications, which need $\mathcal{O}(N_g d_c K)$ complexity. Eq. (24) contains the multiplication operation of three matrices. In view of associative law, it costs $\mathcal{O}(N_d^2 m(m_1 + m_2))$. Let $\mathcal{L}_c$ and $\mathcal{L}_d$ denotes the complexity of GNN_c and GNN_d respectively. $\mathcal{L}_{MLP}$ denotes the complexity of all MLPs. The overall complexity of the proposed methods is $\mathcal{L}_c + \mathcal{L}_d + \mathcal{L}_{MLP} + \mathcal{O}(N_g d_c K + N_g K^2 + N_d^2 m(m_1 + m_2))$.

4. Experiments

In this section, extensive experiments are conducted to verify the effectiveness of the proposed HMM-GDAN for the task of drug response prediction in cancer. The source code is available at <https://github.com/Liuyouf/GNNs>.

4.1. Data preparation

First of all, we need perform data integration for three public cancer-related datasets including GDSC2 (Garnett et al., 2012), CCLE (Barretina et al., 2012) and COSMIC (Tate, Bamford, Jubb, et al., 2019). Then we introduce feature representation methods for drugs and cell lines.

Drugs and cell lines can be found from GDSC2, which directly provides $\ln(\text{IC}_{50})$ values of drug response. **Three omics data arise from CCLE**: gene expression (EXP), solar mutation (MU), copy number variation (CNV). As a matter of fact, there will be 8,2833 available $\ln(\text{IC}_{50})$ values for 580 cell lines and 170 drugs if we do not consider drugs without PubChem ID and cell lines whose partial omics data is missing.

For drugs, we use molecular graph to generate node features and edges as Zhu et al. (2022). The feature is the concatenation of one-hot encoding, binary encoding and integer encoding. The dimension of feature is 77. The details for feature configuration is shown in Table 1.

The 706 cancer-related genes for cell lines are selected from COSMIC (Tate et al., 2019). By Section 3.2, three omics data can form a third-order tensor $C \in \mathbb{R}^{580 \times 706 \times 3}$ by folding operation. Feature matrices for cell lines are $\{C(i, :, :)\}_{i=1}^{580}$, where $C(i, :, :) \in \mathbb{R}^{706 \times 3}$. The gene interaction score can be found in STRING dataset (Szklarczyk et al., 2019). We employ the proposed construction method of multi-view graph in Section 3.2 to generate four views $\{E^{(\text{EXP})}, E^{(\text{CNV})}, E^{(\text{MU})}, E^{(\text{STR})}\}$, where thresholds are empirically set as

$$(\tau^{(\text{EXP})}, \tau^{(\text{CNV})}, \tau^{(\text{MU})}, \tau^{(\text{STR})}) = (0.55, 0.55, 0.30, 0.95).$$

ZINC15 dataset (Sterling & Irwin, 2015) and ChEMBL dataset (Gaulton, Bellis, Bento, et al., 2012) are two large-scale molecular datasets, which are used for node-level self-supervised pretraining with Deep Graph Infomax (DGI) (Velickovic et al., 2019) and graph-level supervised pretraining for drugs.

4.2. Implementation details

Whole dataset is split into 80% training set, 10% validation set and 10% test set and each experiment is repeated with 10 random seeds.

Adam optimizer is adopted in the training. The learning rate, batch size and dropout ratio is selected from the set $\{1e^{-3}, 1e^{-4}, 1e^{-5}\}$, $\{128, 256, 384, 512\}$ and $\{0.2, 0.3, 0.4, 0.5\}$ respectively. In multi-view self-attention graph neural networks for cell lines, the architecture of GNN_c is chosen as GAT (Veličković et al., 2017). The hidden dimension d_c in Eq. (11) is empirically set as 8. The READOUT operator in Eq. (22) is devised as follows. Gralus (Bianchi, Grattarola, & Alippi, 2020) is used to form super nodes for each layer of GAT. We concatenate all representations of super nodes to acquire graph-level representation. In the multi-scale graph neural networks for drugs, the architecture GNN_d is chosen as the modified GIN-GCN. The hidden dimensions m_1 and m_2 are set as 128 uniformly. The hidden dimension in Eq. (26) is also set as 128. Global max pooling is chosen as READOUT operator in Eq. (29). We set $D = 256$ and select q from the set $\{30, 50, 80\}$. Pretraining strategy in Zhu et al. (2022) is followed and performed over ZINC15 dataset (Sterling & Irwin, 2015) and ChEMBL dataset (Gaulton et al., 2012). Similarity augmentation (SA) trick from Zhu et al. (2022) is also adopted in our experiments to further enhance performance.

4.3. Baselines, results and analysis

The chosen comparison baselines are CDRscan (Chang et al., 2018), MOLI (Sharifi-Noghabi et al., 2019), tCNS (Liu et al., 2019), DeepCDR (Liu et al., 2020), GraphDRP (Nguyen et al., 2021) and TGDPR (Zhu et al., 2022). For fair comparison, we replace UGCN in DeepCDR with the branch of drug networks and keep the experimental multi-omics data the same as Zhu et al. (2022). Then the modified method is renamed as DeepCDR*. Following previous works, we use three popular measures to evaluate the performance of methods, namely, root mean square error (RMSE), mean absolute error (MAE) and PCC. For clarity, we introduce these metrics as follows. The RSME, MAE and PCC are three popular metrics for regression tasks. Specifically, RMSE is calculated on the basis of square error (MSE)

$$MSE = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} (\tilde{y}_{ij} - y_{ij})^2. \quad (38)$$

Then

$$RMSE = \sqrt{MSE}. \quad (39)$$

MAE is calculated as

$$MAE = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} |\tilde{y}_{ij} - y_{ij}|, \quad (40)$$

and PCC is

$$\bar{\rho}_{\tilde{\mathbf{y}}\mathbf{y}} = \frac{\sum_{(i,j) \in \mathcal{E}} (y_{ij} - \mu(\mathbf{y}))(\tilde{y}_{ij} - \mu(\tilde{\mathbf{y}}))}{\sqrt{\sum_{(i,j) \in \mathcal{E}} (y_{ij} - \mu(\mathbf{y}))^2} \sqrt{\sum_{(i,j) \in \mathcal{E}} (\tilde{y}_{ij} - \mu(\tilde{\mathbf{y}}))^2}}, \quad (41)$$

where

$$\mu(\mathbf{y}) = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} y_{ij}, \mu(\tilde{\mathbf{y}}) = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \tilde{y}_{ij}. \quad (42)$$

The constructed views are visualized in Fig. 4, which illustrates the self-looped adjacency matrix. Comparison of the setting on input and encoder is shown in Table 2 and experimental results are reported in Table 3.

From Fig. 4, it can be seen that the graph of each view is relatively sparse. This means that only partial important genes will make contribution to the feature extraction and other genes

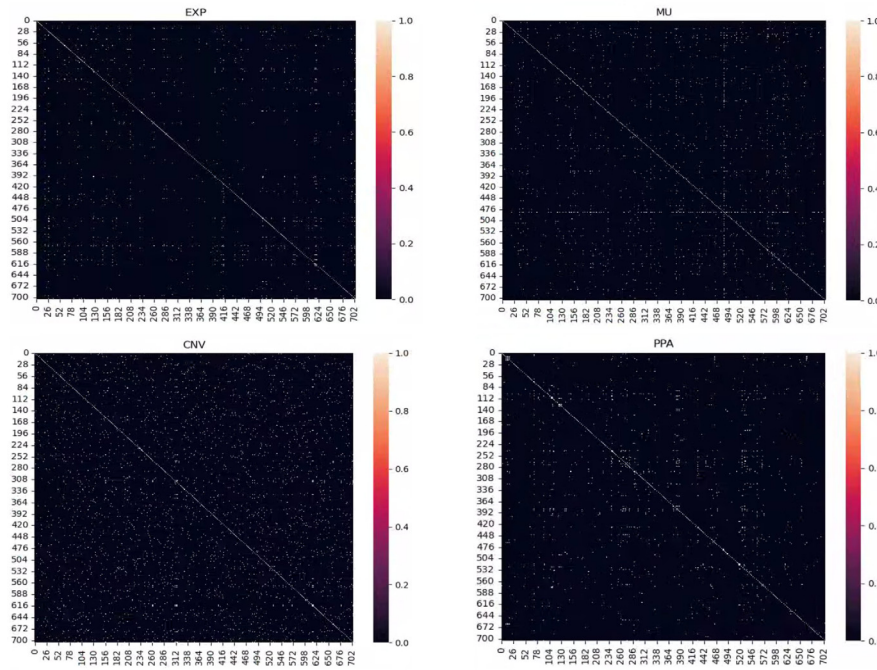


Fig. 4. The visualization of four views. The back point and white point correspond to 0 and 1, respectively.

Table 2
Comparison of the setting on input and encoder.

Approaches	Cell line		Drug	
	Input	Encoder	Input	Encoder
MOLI	Sequence	FCN	–	–
CDRscan	Sequence	CNN	Fingerprint	CNN
tCNNS	Sequence	CNN	SMILES	CNN
GraphDRP	Sequence	CNN	Molecular graph	GNN
DeepCDR*	Sequence	FCN	Molecular graph	GNN
TGDRP _{w/o pre}	Graph	GNN	Molecular graph	GNN
TGDRP	Graph	GNN	Molecular graph	GNN
TGDRP+SA	Graph	GNN	Molecular graph	GNN
HMM-GDAN	Multi-view Graph	Multi-view GNN	Molecular graph	Multi-scale GNN
HMM-GDAN (+SA)	Multi-view Graph	Multi-view GNN	Molecular graph	Multi-scale GNN

Table 3
The results of experiment with respect to three evaluation metrics.

Approaches	RSME	MAE	PCC
MOLI	1.130 ± 0.027	0.875 ± 0.023	0.931 ± 0.005
CDRscan	0.993 ± 0.013	0.737 ± 0.011	0.937 ± 0.002
tCNNS	0.951 ± 0.009	0.700 ± 0.008	0.942 ± 0.001
GraphDRP	0.953 ± 0.020	0.702 ± 0.017	0.942 ± 0.003
DeepCDR*	0.914 ± 0.019	0.674 ± 0.016	0.946 ± 0.002
TGDRP _{w/o pre}	0.894 ± 0.013	0.657 ± 0.010	0.949 ± 0.002
TGDRP	0.885 ± 0.009	0.650 ± 0.006	0.950 ± 0.001
TGDRP+SA	0.877 ± 0.008	0.646 ± 0.006	0.951 ± 0.001
HMM-GDAN	0.865 ± 0.008	0.634 ± 0.005	0.952 ± 0.001
HMM-GDAN+SA	0.853 ± 0.007	0.630 ± 0.005	0.953 ± 0.001

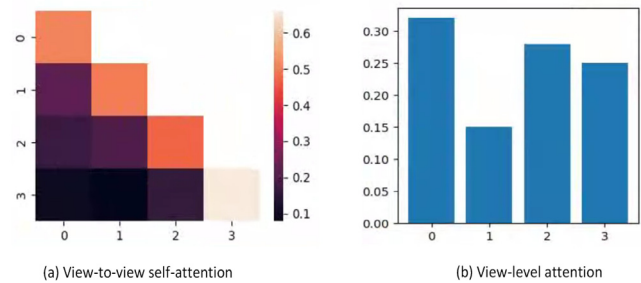


Fig. 5. The learnt attention maps.

keep silent. From Table 2, we announce that it is the first time to construct multi-view graph and devise effective multi-view GNNs for drug response prediction. It can be seen from Table 3 that HMM-GDAN+SA achieves the best performance. This may be attributed to three aspects. The first aspect is that the multi-view graph contains useful information from different views. The collaboration of different views is more informative and may help enhance the expressive power of the proposed method. The baselines are all built on single-view data (sequence or graph) and hence the input information of methods is weaker than ours. The

second aspect may be the design of duplex-attention mechanism, which can help capture the complementary information between views and the importance of each single view. The third respect may be the consideration of multi-scale pattern. The integration of multi-scale features may induce the fusion of features with different meanings so as to achieve stronger representation. Besides, the trick of SA indeed enhances the performance. This may be due to the similarity in drugs and cell lines which induces the hidden relationship among graph-level features such that GNNs can be further applied to generate better representations.

Table 4

The network architecture after ablation.

Cases	Architecture
(a)	HMM-GDAN without multi-view self-attention mechanism
(b)	HMM-GDAN without view-level attention
(c)	HMM-GDAN with Single-scale feature in GCN
(d)	HMM-GDAN with Single-scale feature in GIN
(e)	HMM-GDAN without multi-scale fusion in regressor

Table 5

Results of ablation study.

Cases	RMSE	MAE	PCC
(a)	0.878 ± 0.009	0.640 ± 0.005	0.951 ± 0.001
(b)	0.875 ± 0.010	0.639 ± 0.006	0.951 ± 0.001
(c)	0.884 ± 0.008	0.647 ± 0.005	0.952 ± 0.001
(d)	0.885 ± 0.009	0.647 ± 0.006	0.952 ± 0.001
(e)	0.882 ± 0.008	0.647 ± 0.006	0.951 ± 0.001
HMM-GDAN	0.865 ± 0.008	0.634 ± 0.005	0.952 ± 0.001
HMM-GDAN+SA	0.853 ± 0.007	0.630 ± 0.005	0.953 ± 0.001

For multi-view self-attention and view-level attention, we show the distribution of average learnt attention values in Fig. 5. It can be seen that the self-attention mainly concentrates on the main and secondary diagonal lines. This reveals that the first three views are compatible and they are less complementary with the fourth view. The first view is more important than other views, which means that genes expression data implies more valuable information than others.

4.4. Ablation study

We conduct ablation study for the proposed method from five angles: (a) Remove multi-view self-attention mechanism; (b) Remove view-level attention and just adopt average view-level fusion; (c) Single-scale GNNs with GCN for drugs; (d) Single-scale GNN with GIN for drugs; (e) Remove multi-scale fusion in regressor and only do regression on the feature $[h_i; l_j]$. Architectures after ablation are shown in Table 4 and the experimental results of ablation are exhibited in Table 5.

The results of (a-b) reveal that both multi-view attention and view-level attention have impact on the performance of the proposed method. This may be that multi-view self-attention helps integrate complementary information and view-level attention emphasizes on the importance of each view. The results of (c-e) show that the introduction of multi-scale fusion strategy is favourable. The cause may be that feature of different scales may describe diverse meanings of aggregation styles of GNNs.

5. Parameter sensitivity analysis

Now we analyse the parameter sensitivity of the proposed HMM-GDAN for drug response prediction in cancer. Specifically, we mainly concern the performance change of the proposed method with respect to the learning rate, hidden dimensions and the number of views. For fair comparison, we only adjust the concerned variable and keep other variables unchanged.

The effect of learning rate. From Fig. 6(a), the performance of the proposed method is sensitive to the order of magnitude of learning rate. The large value may not lead the convergence of optimization. The small value may make the optimizer stuck in the bad local minimizers. The best magnitude is 10^{-4} .

The effect of hidden dimensions. From Fig. 6(b), we found that the hidden dimension q in Eq. (34) has impact on the performance of the proposed method. Specifically, when it takes small values, the performance degrades. The cause is that small dimension gives rise to the information loss. If it takes large values, information redundancy will occur in the resulting representations.

Empirically, the best chosen interval may be [80, 100]. Other hidden dimensions do not produce significance influence on the performance, hence we just report the result of performance with respect to q .

The influence of the number of views. From Fig. 7, the performance of the proposed method generally increases as the increment of the number of views. This may be that more views will embody more prior knowledge hidden in data, which is beneficial for the learning process and generalization power. We also found that the performance is sensitive to the fourth view when the number of views is fixed. The first three views is built on multi-omics data and the fourth view comes from STRING dataset. This sensitivity may be caused by the heterogeneity between first three views and the fourth view.

6. Discussion

The effectiveness of the proposed HMM-GDAN has been verified in the experiments. Multi-view graph learning scheme indeed makes progress in the prediction of drug response prediction. However, there is still some problems worthy studying in the future. For example, the multi-view graph for drugs is difficult to construct. Because the encoding of nodes in molecular graph is mainly one-hot format, the construction method in Section 3.2 may not work effectively. It is desirable to develop new methods to perform multi-view GNNs for drugs. Besides, the running efficiency is also a realistic problem. The running time comparison among selected methods is shown in Fig. 8. Since multi-view graph is used in the proposed methods, the running time is not advantageous. There is still room for improvement on efficiency in the future.

7. Conclusion

In this paper, we propose hybrid multi-view and multi-scale graph duplex-attention networks for drug response prediction in cancer. The construction method of multi-view graph is provided, which is an important ingredient for the proposed method. Multi-view self-attention mechanism and view-level attention mechanism work together, which take the complementary information among views into account and emphasize on the importance of views collaboratively. Multi-scale fusion in both GNNs and regressor is established. Extensive experiments demonstrate the effectiveness of the proposed method when compared with the state-of-the-art ones by means of commonly used regression measurements.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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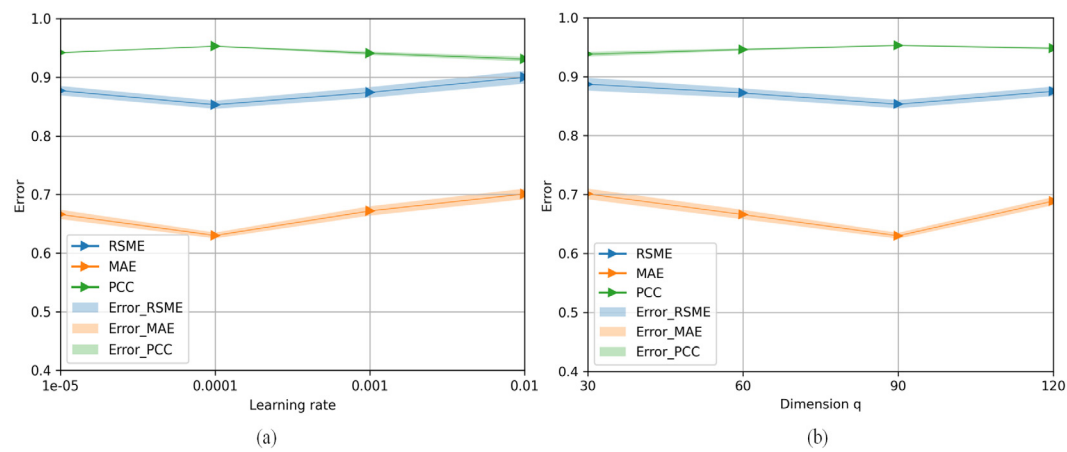


Fig. 6. The sensitivity regrading learning rate and hidden dimensions.

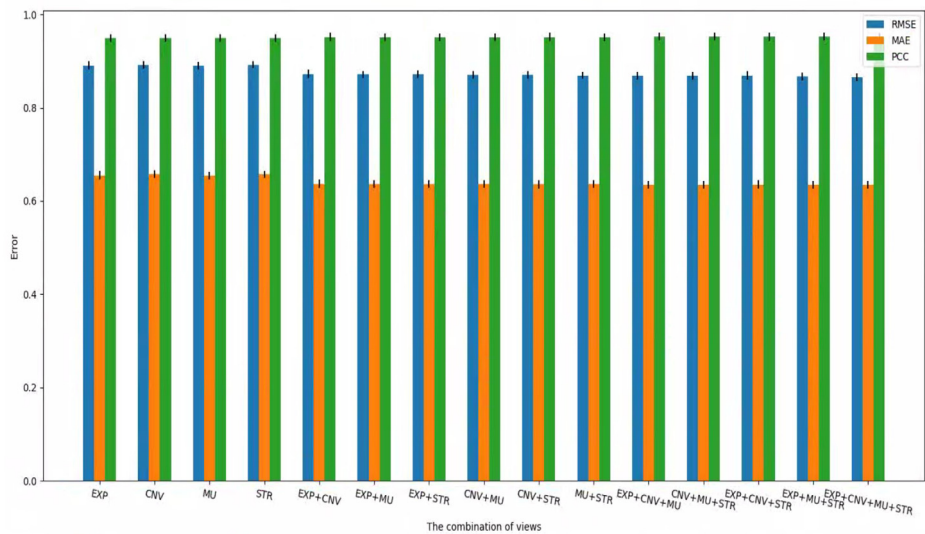


Fig. 7. The sensitivity of the proposed method with respect to the number of views.

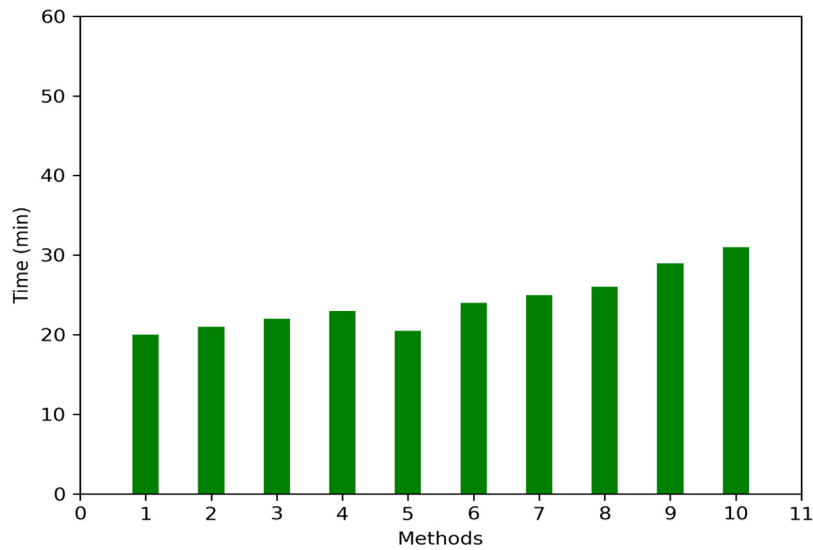


Fig. 8. The comparison of running time. The number in x-axis represents the methods in Table 3 from top to down.

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